

MCMaster ENGINEERING SOCIETY LARGE EVENTS MANAGEMENT SYSTEM

Team Members

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Overview

MacSync is a centralized event management platform that unifies registration, ticketing, waivers, and check-in into one system.

It provides:

- A mobile-first app for students
- A web dashboard for organizers
- A seamless, efficient event experience

Key Features



Secure ticket registration with real-time seat reservation, lockout, and concurrency control



Table and bus signups with capacity tracking and enforced booking limits



Stripe payment integration with checkout sessions, payment logging, and refund handling



QR code tickets for fast check-in, validation, and fraud prevention at events



Admin dashboard with event management, custom form builder, and role-based access control



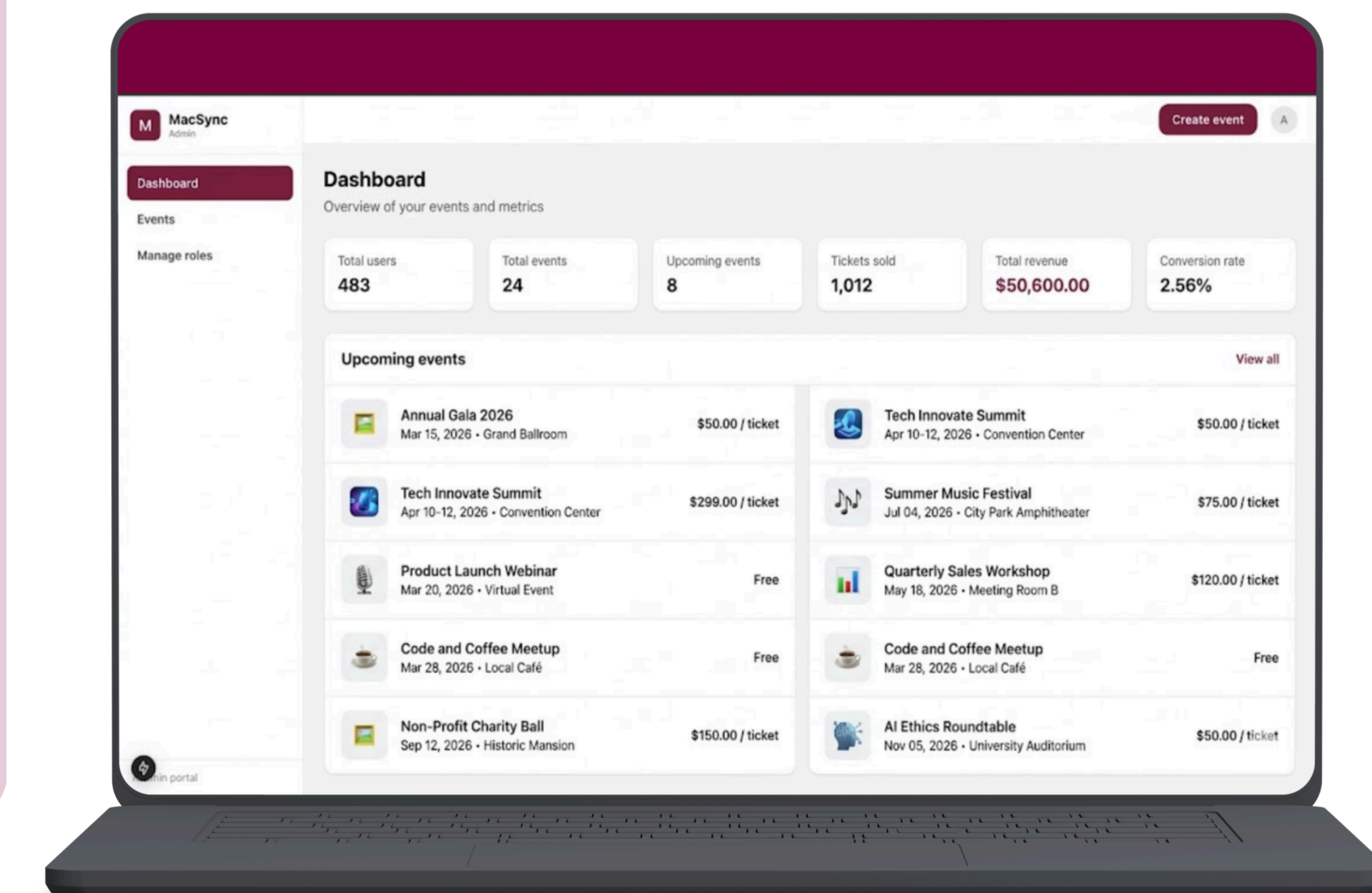
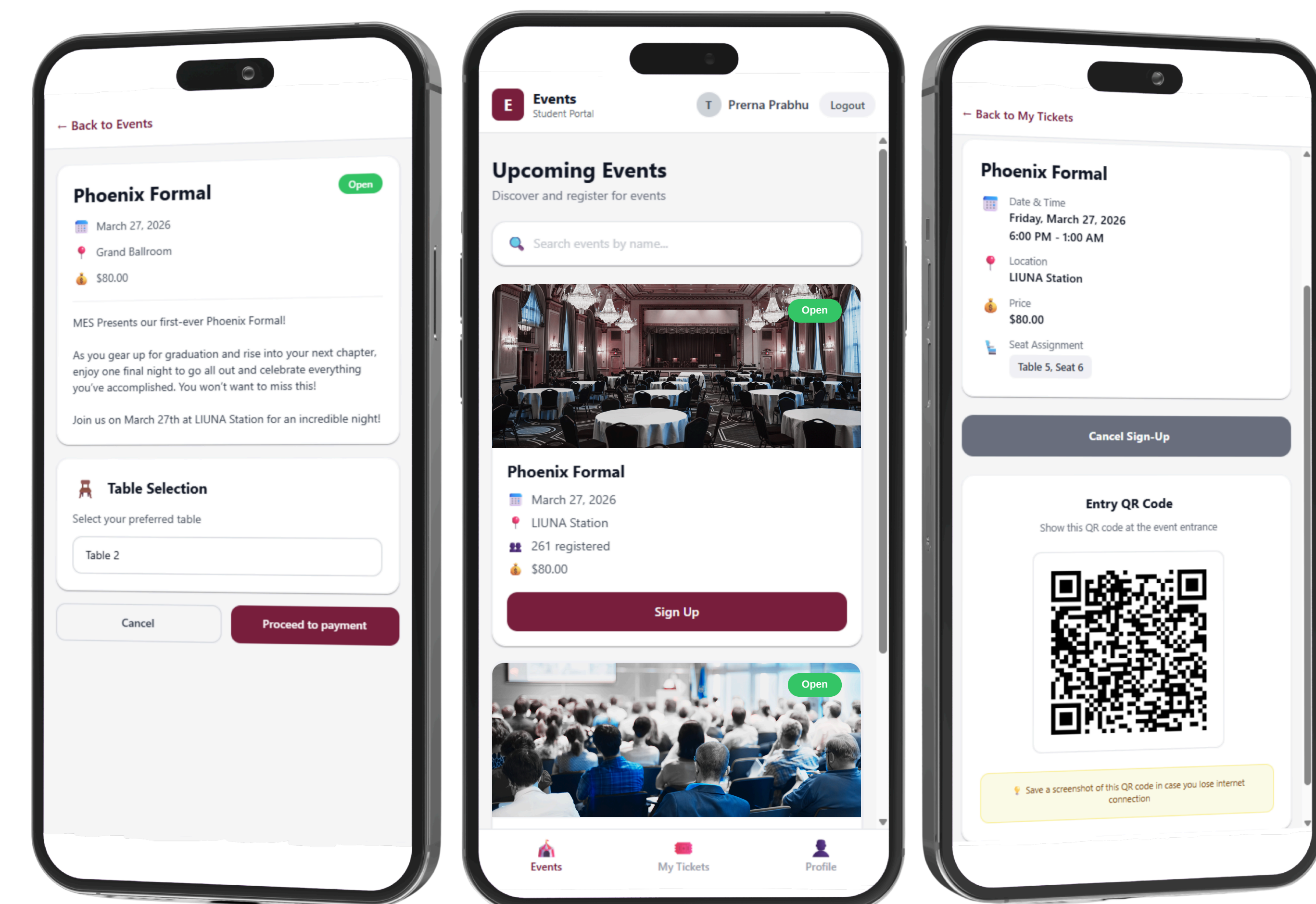
Push notifications, reminders, and analytics dashboards for engagement, attendance, and sales insights



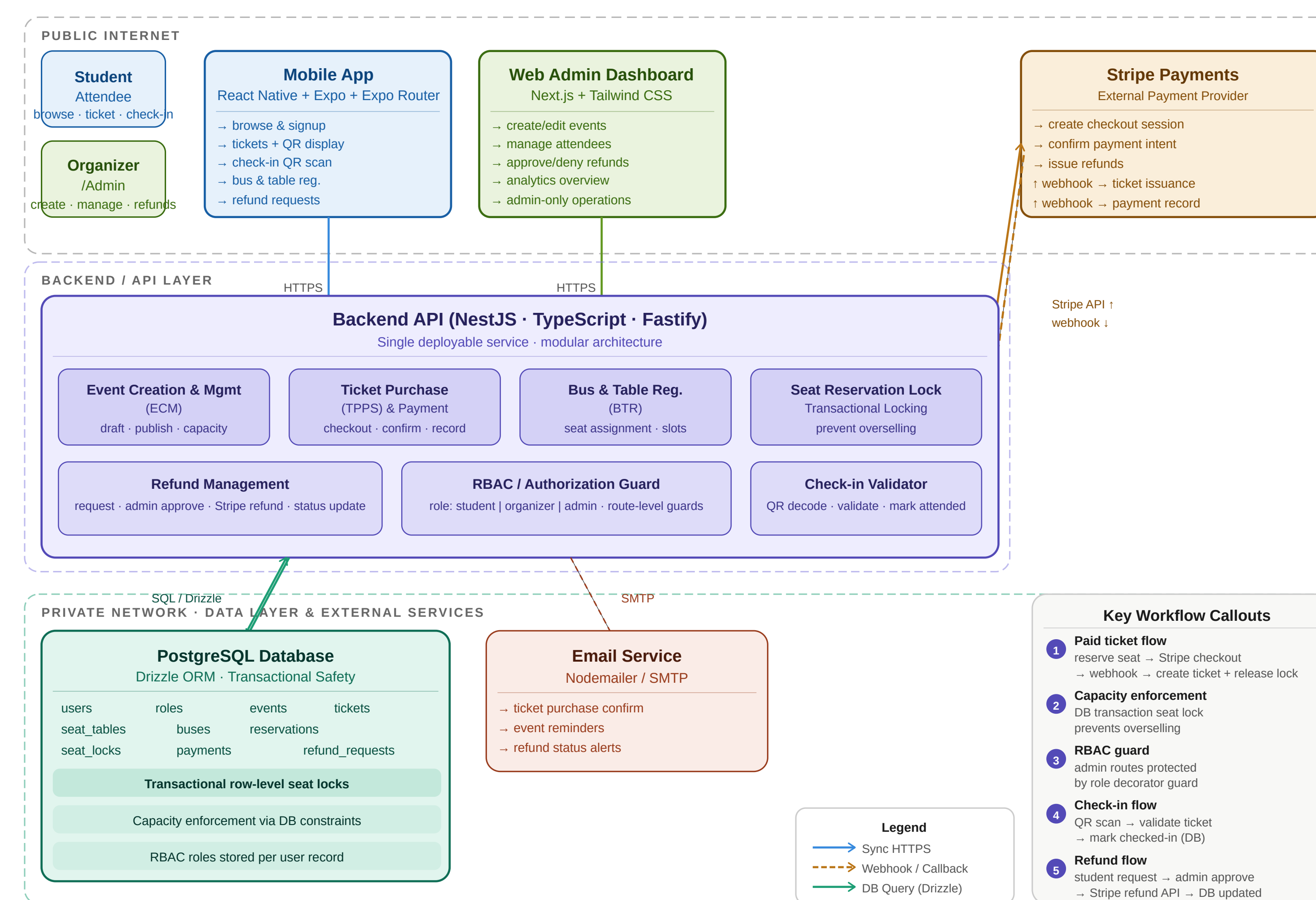
Attendee management with user profiles, dietary restrictions, and digital waiver submission

Problem

- Event management is fragmented across multiple platforms (Google Forms, Spreadsheets, Instagram)
- Students struggle with accessing info, registering, and receiving updates
- Organizers lack centralized tools for ticketing, attendee tracking, and logistics
- Results in inefficiency, confusion, and wasted volunteer time



System Architecture



Challenges & Solutions

Concurrent Booking: Prevented double-booking using PostgreSQL row-level locking

Stripe Integration: Implemented secure checkout, webhooks, and refunds

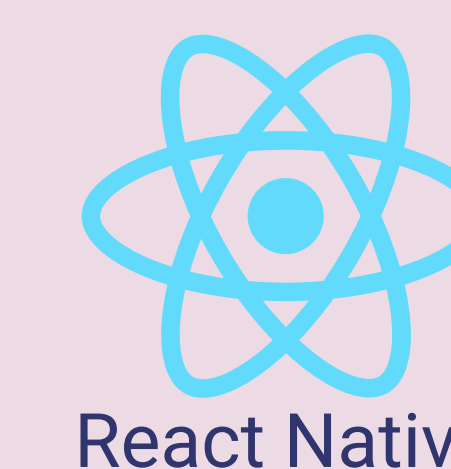
Multi-Platform System: Unified mobile + web via single NestJS API

Scalability: Designed to support hundreds of concurrent users

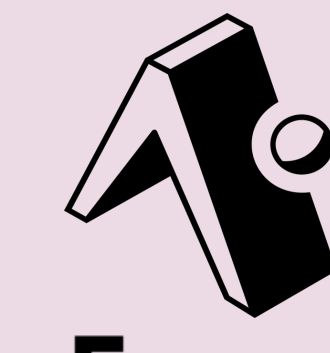
Metrics & Impact

- Built to support 300+ attendees per event
- Real-time seat reservation with zero double-booking
- Secure end-to-end payment processing via Stripe
- Mobile-first experience improves accessibility and engagement
- Reduced manual coordination for organizers
- Replaced 4+ fragmented tools with one centralized system

Tech Stack



React Native



Expo



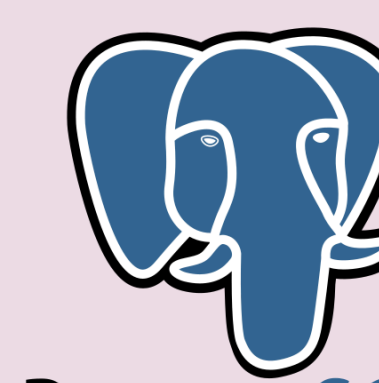
Next.js



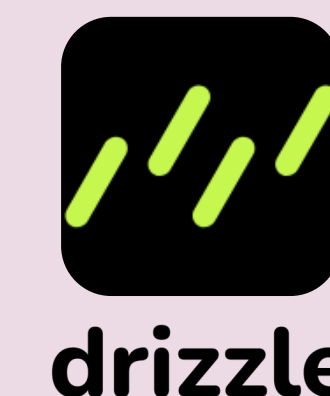
fastify



nest



PostgreSQL



drizzle



Tailwind CSS



Jest